CENG381 Stochastic Processes

Stationary Distribution — Practice 1

Q1 (30 points). A marine biologist tags fish that move between two coastal areas, A and B. Each day, 2 out of every 3 fish in area A swim to area B, while only 1 out of every 5 fish in area B return to area A.

a) Model this as a two-state Markov chain and write the transition matrix P (states ordered A, B).

b) Find the stationary distribution $\pi = (\pi_A, \pi_B)$ by solving $\pi \cdot P = \pi$ together with $\pi_A + \pi_B = 1$. Set up the balance equations explicitly.

c) Interpret π_A: in the long run, what fraction of days does a tagged fish spend in area A?

Q2 (35 points). A factory machine cycles through three states each day: Working (W), Under Maintenance (M), and Failed (F). The transition matrix is:

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        W      M      F
W  [ 1/2    1/3    1/6 ]
M  [ 1/2     0     1/2 ]
F  [  1      0      0  ]
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a) Find the stationary distribution $\pi = (\pi_W, \pi_M, \pi_F)$ by solving the balance equations $\pi \cdot P = \pi$ with $\pi_W+\pi_M+\pi_F = 1$.

b) Verify your answer by computing $\pi \cdot P$ directly and confirming each component equals the corresponding π value.

c) Test whether the chain is time-reversible by checking the detailed balance condition $\pi_i \cdot P_{ij} = \pi_j \cdot P_{ji}$ for the pair (W, M) and the pair (W, F). What do you conclude?

Q3 (35 points). An ATM machine is classified at the start of each business day as: Full (F) — cash above 80%, Normal (N) — cash between 20–80%, or Low (L) — cash below 20%. The transition probabilities are:

- From F: stays Full with prob 1/2, drops to Normal with prob 1/2.
- From N: refilled to Full with prob 1/4, stays Normal with prob 1/2, drops to Low with prob 1/4.
- From L: always refilled to Normal overnight.

a) Write the 3×3 transition matrix P (rows and columns ordered F, N, L).

b) Find the stationary distribution $\pi = (\pi_F, \pi_N, \pi_L)$.

c) What fraction of days is the ATM in the Low state? Compare π_F and π_L. Which state is most frequently visited in the long run?